

Design Decisions Document

C++ OpenGL 3D Kitchen Scene

Portfolio Project - Computational Graphics and Visualization

Project Summary

For this project, I created a 3D kitchen scene using C++ and OpenGL. The scene includes kitchen objects such as stools, an island, countertops, and additional structural details. The goal of the project was to create a fully realized 3D world while demonstrating skills in shape creation, object placement, texturing, lighting, projection modes, and camera movement.

My main design goal was to make the kitchen feel organized, recognizable, and visually balanced. I wanted the scene to look like a believable interior space rather than a collection of separate objects, so I focused on how each object related to the surrounding environment.

Scene Concept and Layout Decisions

I chose a kitchen scene because it allowed me to work with a variety of object types and surfaces. A kitchen naturally includes large fixed objects, repeated smaller objects, flat surfaces, and details that benefit from lighting and texture. This made it a good environment for practicing computational graphics concepts.

Scene Element	Design Decision
Kitchen island	Used as the central object in the scene to organize the layout and provide a clear focal point.
Stools	Placed around the island to make the scene feel functional and lived-in. Repetition helped create balance.
Countertops	Added as recognizable kitchen surfaces and used to strengthen the interior environment.
Additional objects and surfaces	Included to make the scene feel more complete and to demonstrate object positioning and scaling.

The layout was built with object relationships in mind. I adjusted the position and scale of objects so the kitchen would appear proportional. I used transformations to control where objects appeared in the world and to prevent the scene from looking crowded or disconnected.

Shape and Object Construction Decisions

The scene was developed by manipulating basic 3D shapes and transforming them into recognizable kitchen objects. I used scaling, translation, and rotation to create objects with the correct proportions and placement. This approach allowed me to build more complex objects from simpler geometry.

For example, the island and countertops were created from larger rectangular forms, while the stools required more careful placement and repeated object construction. By reusing and adjusting shapes, I was able to create consistency throughout the scene while still giving each object a clear purpose.

Texture Decisions

Texturing was used to make the objects look less flat and more realistic. I applied textures to surfaces so the kitchen would have visual variety and so different objects would be easier to identify. Texture choices helped communicate material differences between surfaces such as counters, seating, and other kitchen elements.

I also considered how textures interacted with lighting. A texture that looked acceptable without lighting could appear too dark, too bright, or unclear once lighting was added. Because of this, I tested textures in the actual scene and made adjustments based on how they appeared during rendering.

Lighting Decisions

Lighting was an important part of making the kitchen scene feel three-dimensional. I added lighting so that objects would have depth, highlights, and shadows instead of appearing flat. The lighting helped separate objects from the background and made the scene easier to view.

My lighting decisions focused on visibility and realism. I wanted the scene to be bright enough for the viewer to understand the layout, while still showing how light affects different surfaces. Adjusting the light position and intensity helped improve the overall appearance of the scene.

Camera and Projection Decisions

Camera movement was added so the user could navigate the 3D kitchen scene. This made the project more interactive and allowed the scene to be viewed from multiple angles. Adding camera controls also helped me test whether the objects looked correct from different perspectives.

I also worked with different camera operations, including orthographic and perspective views. The orthographic view was useful for viewing objects without perspective distortion, while the perspective view made the scene feel more natural and realistic. Comparing these views helped me better understand how projection choices affect a 3D environment.

Development and Iteration Decisions

The project was developed through iteration. I did not complete the scene in one pass. Instead, I built the basic scene first, tested it, and then made improvements. Each milestone helped me add new features and refine the work I had already completed.

Iteration was especially important when adjusting object positions, lighting, camera settings, and textures. Many objects needed to be moved, resized, or modified after viewing the scene in OpenGL. This process helped me learn that visual programming requires frequent testing because small code changes can have a major effect on the final image.

Major Technical Decisions

Area	Decision	Reason
Programming language	Used C++ with OpenGL	This supported low-level graphics programming and direct control over rendering.
Object creation	Built objects from basic shapes	This made it possible to construct a complete scene while practicing transformations.

Transformations	Used scaling, translation, and rotation	These operations allowed each object to be placed and sized correctly.
Textures	Applied image-based surface detail	Textures improved realism and helped distinguish materials.
Lighting	Added scene lighting	Lighting created depth and improved the visual quality of the scene.
Camera	Added movement and projection changes	Camera controls made the scene interactive and easier to inspect.

Challenges and Solutions

One challenge was getting objects to appear in the correct location and scale. In a 3D scene, small changes to transformation values can make an object appear too large, too small, or out of place. I solved this by testing frequently and adjusting the values until the scene looked balanced.

Another challenge was balancing lighting and texture. Lighting can change how textures appear, so I had to view the scene repeatedly and make adjustments. Camera movement also required testing because the scene needed to look acceptable from more than one angle.

Final Design Reflection

My design decisions were focused on creating a realistic and understandable 3D kitchen environment. I used a kitchen island as the main focal point, added stools and countertops to make the scene recognizable, and used textures and lighting to improve visual quality. Camera movement and projection changes helped make the project more interactive and demonstrated my understanding of how users view and navigate 3D spaces.

This project helped me develop stronger skills in computational graphics, visual design, and iterative programming. I learned how object placement, transformations, lighting, textures, and camera systems work together to create a complete 3D scene.